Chanyeok Choi

Robot Engineer · MS in Applied AI, Hanyang University

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Profile

Robot engineer working on **self-supervised critical phase detection** for vision-language-action policy refinement (preprint, under review). Broader interests&58 safe and generalizable robot learning, multi-agent reinforcement learning, and sim-to-real transfer. Previously built field-deployed service robots (auto-climbing mopping platform, autonomous delivery platform) — most of my intuition about real-world control and edge cases comes from there.

Education

Hanyang University, Applied Artificial Intelligence — Seoul, South Korea, Feb 2022 – present

- Advisor: Prof. Youngmoon Lee
- Research on safety-RL, poisoning attacks on multi-agent RL, and vision-language-action policy failure detection.

Hanyang University, Robotics — Seoul, South Korea, Mar 2018 – Feb 2022

- Advisors: Prof. Youngmoon Lee, Prof. Taejoon Park, Prof. Changsoo Han

Experience

Hanyang University, Graduate Research Assistant — Seoul, South Korea, Mar 2022 – present, 4 years 3 months

Robot learning research under Prof. Youngmoon Lee.

- **Self-Supervised Critical Phase Detection (CPD)**: step-level premature-commitment detector and ≈ 1K-parameter residual adapter for frozen π_0 / $\pi_{0.5}$ VLA policies on LIBERO; cross-modal execution gap between action-expert latent and proprioception (preprint, under review).
- **Poisoning attacks on multi-agent RL**: adversarial reward-poisoning study on multi-agent control, including taxi ride-sharing systems.
- **Safety-RL**: bridging the sim-to-real gap by allowing safe optimization and exploration during robot learning.
- **Vision (HumanPose)**: hybrid PoseNet/SegNet approach for crowded scenes with overlap and occlusion.

KITECH — Process Platform Research Division, Research Intern — South Korea, Mar 2021 – Aug 2021, 6 months

- Researched AI methods for detecting defective products on high-speed conveyor belts.

Hanyang University — CAI Lab, Research Intern — Seoul, South Korea, July 2019 – Feb 2020, 8 months

- IKEA Furniture Assembly Collaborative Robot: RL simulation in which two manipulators identify randomly placed furniture parts and assemble them collaboratively.

Publications

Self-Supervised Critical Phase Detection for VLA Refinement
Self-supervised step-level critical-phase detector and ≈ 1K-parameter residual adapter that refines frozen π_0 / $\pi_{0.5}$ VLA policies inside detected phases.
Chanyeok Choi, Youngmoon Lee
angledsugar.github.io/projects/0_cpd

Poisoning Attacks on Multi-Agent Reinforcement Learning Systems
Chanyeok Choi, Jaehwan Cho, Youngmoon Lee

HAPtics: Human Action Prediction in Real-time via Pose Kinematics
Niaz Ahmad, Saif Ullah, Jawad Khan, Chanyeok Choi, Youngmoon Lee

Causes and Fixes of Unexpected Drone Shutoffs
Hojun Choi, Chanyeok Choi, Youngmoon Lee

Snapbot: Enabling Dynamic Human Robot Interactions for Real-Time Computational Photography
Chanyeok Choi, Youngmoon Lee

Leveraging Keypoints as Dynamic Centroids for Unified Representation of Human Pose and Instance Segmentation
Niaz Ahmad, Jawad Khan, Chanyeok Choi, Youngmoon Lee, Kang G. Shin

SSK-DNN: Semantic and Sentiment Knowledge for Incremental Text Sentiment Classification
Jawad Khan, Niaz Ahmad, Chanyeok Choi, Saif Ullah, Gyu Rin Kim, Youngmoon Lee

Awards

Information & Communication Planning and Evaluation President Award (2nd place) — SW Talent Festival Jan 2023
AnimeGAN Filter Photographer&58 a manipulator that frames and captures stylized portraits under safety-RL constraints.

Capstone Design Award (3rd place) — Hanyang University Jan 2021
Guide Robot Dog&58 a full-stack quadruped robot motivated by guide dogs, intended to assist visually impaired pedestrians.

Teaching

Lead Teaching Assistant — Robot Vision System: Hanyang University, Fall 2022 & Fall 2023. Lead other graduate TAs for a class of 30–40 students; developed an autonomous robot training and vision environment in Unity. [Materials]

Teaching Assistant — Robot Learning: Hanyang University, Spring 2022 & Spring 2023. Class of 40–50 students. Designed a Pac-Man RL project and an ultrasonic-sensor self-driving car kit for real-world maze escape. [Materials]

Skills

Robot Learning

Perception

Engineering

Languages

Korean
Native speaker

English
Professional working proficiency

Interests

Research Interests